iMouse: Eye Controlled Mouse

Introduction

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- **Goal:** Control the mouse hands-free using a standard webcam.
- **Inputs:** Head movement, eye position and facial gestures.
- **Actions:** Cursor movement, clicks, scrolling and pause/resume.

TEAM

Akshat Singhania (230102)
Deep Aryan Singh (230346)
Harsh Pratap Singh (230447)
Rohit Vinod Atkurkar (230872)

Problem Statement

iMouse: Eye Controlled Mouse

- **Cursor control gap:** Cursor following alone leaves clicking, scrolling and pausing without hands-free control.

OBJECTIVE

Movement

Hands-free cursor movement

Clicks

Left and right clicks

Scrolling

Bidirectional scrolling

Pause / resume

Pause and resume control

- **Setup:** One RGB webcam, one visible face and short neutral calibration; no dedicated eye-tracking hardware.
- **System latency:** Reduce avoidable mouse-library delays and stale frames.

METHODOLOGY · SYSTEM DESIGN

How We Approached the Problem

1

Capture & landmarks

OpenCV captures video; MediaPipe Face Mesh estimates facial landmarks.

2

Neutral calibration

45 consecutive valid frames establish neutral iris position, open-eye reference and head roll.

3

Gesture separation

Calibrated per-eye EAR and a temporal state machine separate winks from blinks. Clicks are confirmed on release; the cursor is held during the gesture.

4

Smoothing & speed

One Euro filter stabilizes movement. Process the latest frame and remove PyAutoGUI's default 100 ms pause while retaining its fail-safe.

Contribution: Integrate these techniques into one everyday desktop controller.

INTEGRATION · WHAT SETS IT APART

One webcam, one desktop controller

1

Standard hardware

Only an ordinary webcam is needed. No eye tracker, headset or wearable sensor.

2

Wink vs. blink

Per-user calibration of eye aspect ratio and a temporal state machine tell deliberate winks from natural blinks.

3

Smooth, current cursor

One Euro filtering stabilizes the pointer, and each frame processed is the latest one, so control stays responsive.

4

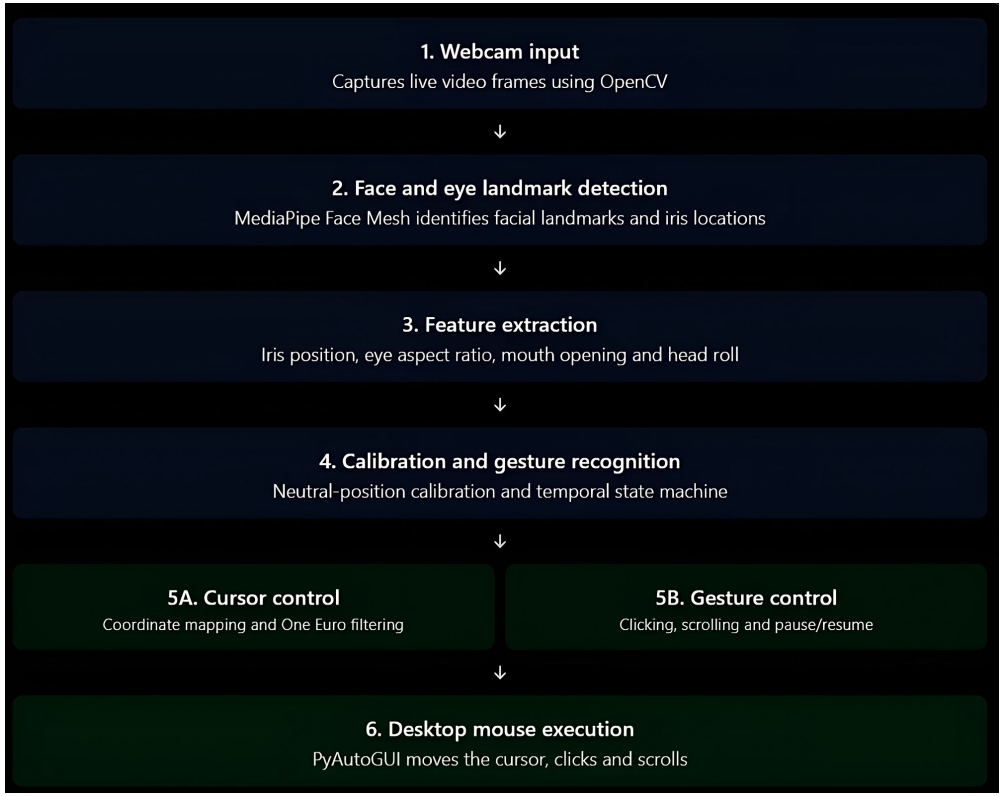
Several desktop actions

Cursor movement, clicks, scrolling and pause/resume all run through one gesture pipeline into PyAutoGUI.

Contribution: Integrating established techniques into one everyday desktop controller; no claim of a new method.

ARCHITECTURE · END-TO-END FLOW

Our Implemented Pipeline



Tracking Stream

One Euro Filter smooths neutral-relative iris displacement mapped to screen coordinates.

Gesture State Machine

Winks versus blinks; head tilt scrolls; a 0.6s mouth hold toggles pause/resume.

OS Event Dispatch

PyAutoGUI moves, clicks and scrolls with fail-safe limits intact.

Live Code Demonstration

Code map

- imouse/core.py — EAR, gesture state machine and One Euro filtering
- imouse/capture.py — threaded OpenCV latest-frame capture
- imouse/app.py — tracking and PyAutoGUI desktop events
- tools/target_practice.py — interactive target-selection demo

Demo sequence

1. Run python mouse-controller.py --dry-run; show tracking and 45-frame neutral calibration.
2. Run python mouse-controller.py; blink normally, then wink for left and right clicks.
3. Tilt head to scroll; hold mouth open 0.6 seconds to toggle pause/resume.
4. Run python tools/target_practice.py and select targets with wink clicks.

Limitations and Future Improvements

Current Limitations

- Head translation shifts iris coordinates relative to the calibrated neutral center, affecting the cursor.
- Lighting, thick-rimmed glasses, occlusion and sharp head rotations can degrade landmarks.
- Independent, repetitive winks can be difficult or tiring.
- Verification relies on synthetic landmark tests and offline replay, not large-scale live trials.

Future Improvements

- Compare original and revised iMouse in live target-selection trials using Fitts' Law benchmarks.
- Measure false clicks, missed gestures, cursor jitter and target acquisition time.
- Test customizable dwell-to-click without conscious facial gestures.
- Explore calibration and real-time 3D pose vectors to decouple head pose from gaze.

FOUNDATIONS

Research and References

Source / Paper	Relevance to iMouse
Iacobelli et al. (2023)	Classical iris localization; low-cost baseline
MediaPipe Face Mesh & Iris	Pretrained landmarks for feature extraction
Soukupová & Čech (2016)	EAR geometry; wink vs. blink classification
Casiez et al. (2012) — One Euro Filter	Speed-adaptive smoothing of cursor movement
OpenCV VideoCapture & PyAutoGUI Docs	Capture, desktop input, pause and fail-safe
Original iMouse baseline	Historical baseline
NASA/scikit-image astronaut image	Offline smoke-test

Thank You

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[https://github.com/harshpsi9194/EE604_Course-Project.git]